Session 9. Model scenario analysis

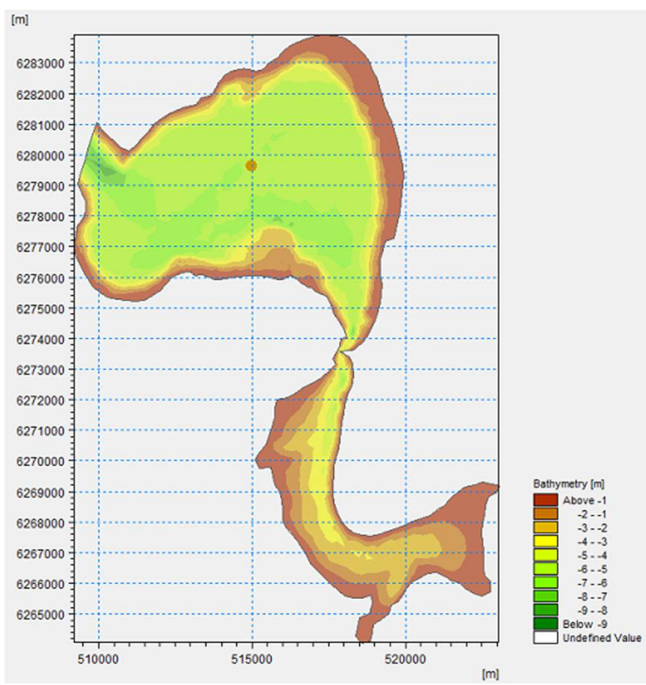
In this session, you will work with the output from the Skive Fjord model scenarios that were discussed during Session 8. Your task is to explore, interpret, and discuss the ecological implications of the scenario results in a management context. The scenario designs can vary from the original idea due to time constraints in executing them. You can find visualizations on how the scenario forcing deviates from the baseline forcing at the end of this exercise.

The baseline and scenarios executed for this exercise:

Scenario	Description	Forcing changes			
		Variable	Spatial	Temporal	Magnitude
Baseline	Baseline	No changes			
1	Summer nutrient increase[a]	Nutrients*	Streams (sources)	May to Sep.	+20 %
2	Summer nutrient decrease[b]	Nutrients*	Streams (sources)	May to Sep.	-100 %

[a] Resembling increased precipitation resulting in increased nutrient run-off; [b] Only summer reduction, no winter increase; * Nutrients: NO_3, NH_4, PO_4, N and P in detritus (DN and DP), colored and labile dissolved organic matter (CDON, CDOP, LDON, LDON) and phosphate adsorbed to inorganic matter (IPss).

Model domain is constrained to Lovns Broad and Hjarbæk Fjord with surface and bottom water quality variables extracted from the local NOVANA station in the center of Lovns Broad. Surface and bottom model results are extracted at 1 m and 6 m depth, respectively.



Exercise

1. **Familiarize yourself with the scenario outputs**
   Investigate the baseline and scenario output files. Review the structure of the files and identify the variables relevant in a eutrophication management context.

2. **Scenario analysis**
   Examine how the scenario affects the ecosystem compared to the baseline. Consider:
   a. Which components respond most strongly?
   b. What processes likely drive the changes in water quality?
   c. Do the results support or contradict your original hypothesis?

   Create 2–3 visualizations (e.g., plots, difference curves, maps) that illustrate the key outcomes.

   Please consider:
   How will you best visualize the potential difference between baseline and your scenario, and how to visualize the mechanism(s) driving this change?
   What comparisons to the baseline run are necessary to understand the scenario's impact?

3. **Group discussion**
   We will wrap up this session with a discussion on the scenario results. Consider:
   a. the ecological and management relevance of the results,
   b. how confident you are in the patterns observed,
   c. any surprising behaviors or uncertainties in the model response.

Changes to nutrient concentration in inlets to Lovns Broad

The two nutrient scenarios were implemented to all streams flowing into Lovns Broad and Hjarbæk Fjord.

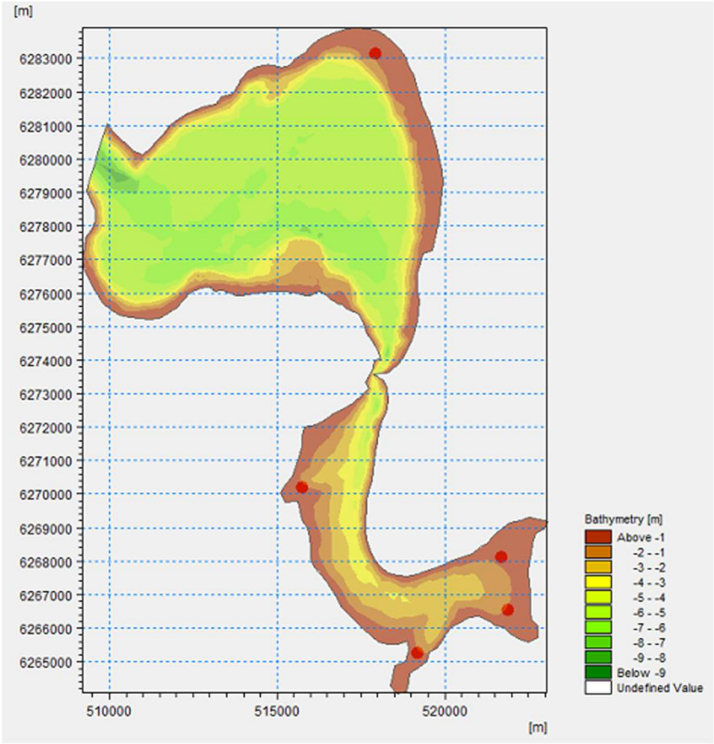
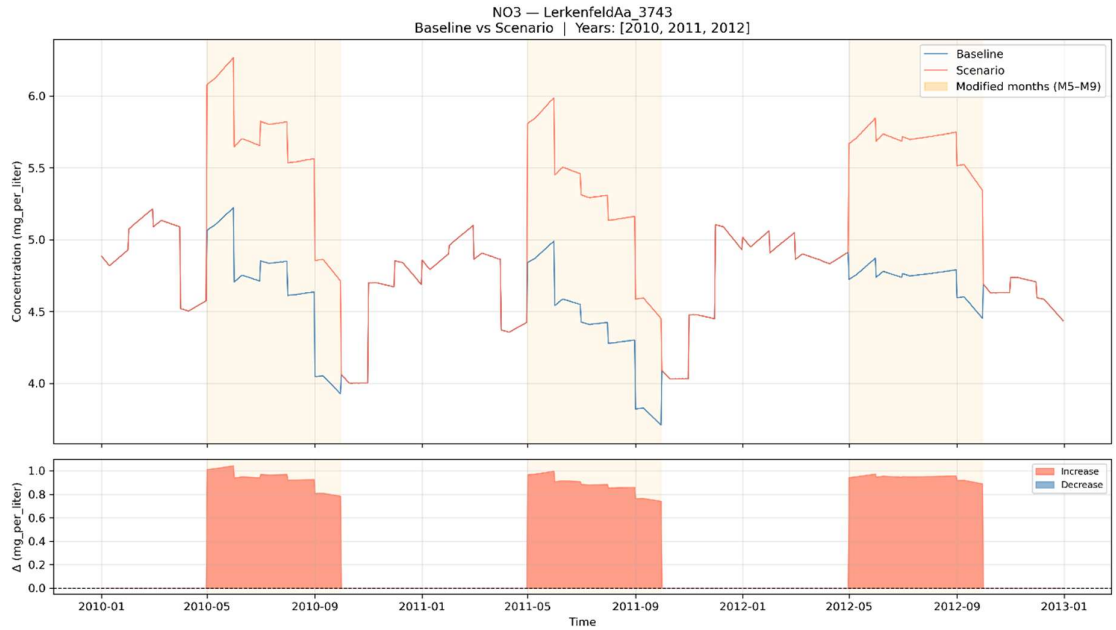
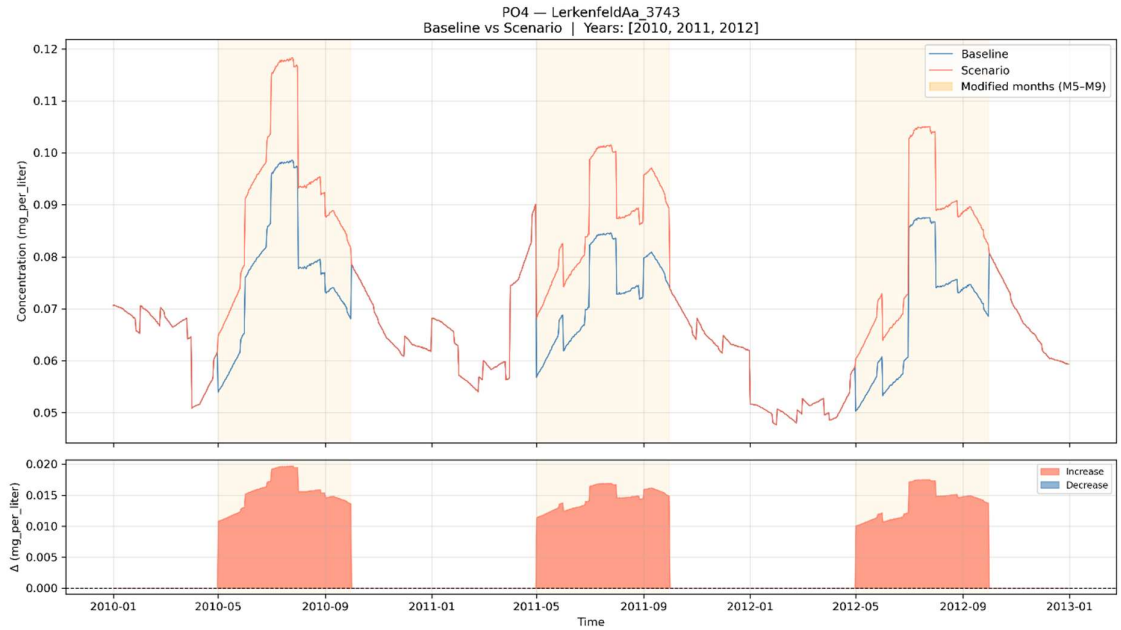


Figure 1. Bathymetry of model domain, Lovns Broad and Hjarbæk Fjord, with sources (streams, red dots).

Scenario 1: 20% increase in summer N and P stream concentrations

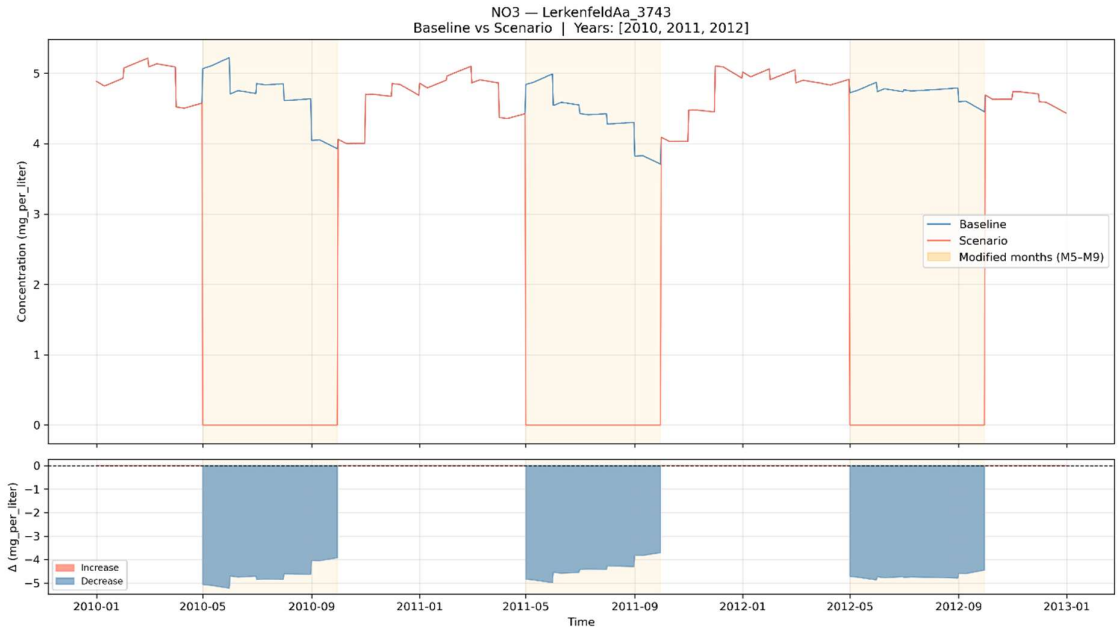
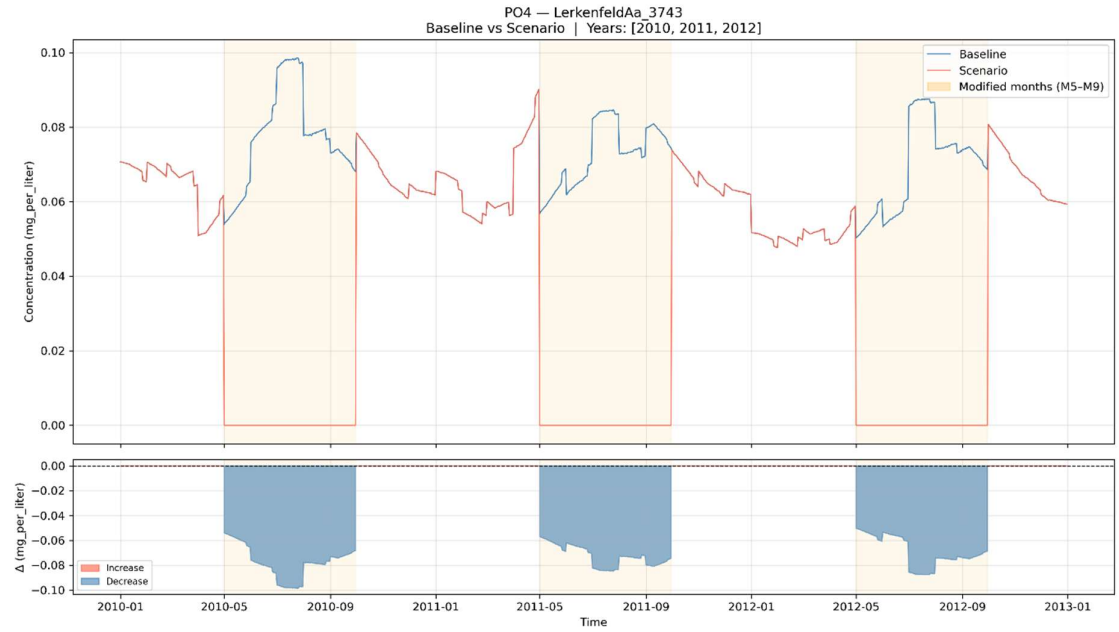
Examples of changes to nutrient concentration for scenario 1:





Scenario 2: 100% reduction in summer N and P stream concentrations

Examples of changes to nutrient concentration for scenario 2: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